

SVM

$$\min_{w, b} \underbrace{\frac{1}{2} \|w\|^2}_{\text{regularizer}} + C \underbrace{\sum_{n=1}^N \max\{0, 1 - y_n (\langle w, x_n \rangle + b)\}}_{\text{error term.}}$$

hinge loss

$$l(t) = \begin{cases} 0 & \text{if } t \geq 1 \\ 1-t & \text{if } t < 1 \end{cases}$$

minimize hinge loss

$$\min_t \max\{0, 1-t\} \Rightarrow \begin{matrix} \min_{\xi, t} \xi \\ \text{s.t. } \xi \geq 0, \xi \geq 1-t \end{matrix}$$

$$L_{\text{hinge}}(w, b, x_i, y_i) = \max(0, 1 - y_i (w^T x_i + b))$$

Ridge Penalty

$$L_{\text{ridge}}(w) = \frac{1}{2} \|w\|^2$$

$$J_{\text{ERM}}(w, b) = \frac{1}{2} \|w\|^2 + C \sum_{i=1}^N \max(0, 1 - y_i (w^T x_i + b))$$

Duality with Lagrange Multipliers:

$$(1) \mathcal{L}(w, b, \xi, \alpha, \gamma) = \frac{1}{2} \|w\|^2 + C \sum_{n=1}^N \xi_n - \sum_{n=1}^N \alpha_n (y_n (\langle w, x_n \rangle + b - 1 + \xi_n)) - \sum_{n=1}^N \gamma_n \xi_n$$

$$\left. \frac{\partial \mathcal{L}}{\partial w} = w^T - \sum_{n=1}^N \alpha_n y_n x_n^T \right\} = 0 \Rightarrow w = \sum_{n=1}^N \alpha_n y_n x_n$$

$$\left. \frac{\partial \mathcal{L}}{\partial b} = - \sum_{n=1}^N \alpha_n y_n \right\} = 0 \Rightarrow \sum_{n=1}^N y_n \alpha_n = 0 \quad \left(\begin{array}{l} \text{from} \\ \text{with } b \\ \text{variable} \end{array} \right)$$

$$\left. \frac{\partial \mathcal{L}}{\partial \xi_n} = C - \alpha_n - \gamma_n \right\} = 0 \Rightarrow \sum_{i=1}^N (C - \alpha_n - \gamma_n) = 0$$

$$(2) D(\xi, \alpha, \gamma) = \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N y_i y_j \alpha_i \alpha_j \langle x_i, x_j \rangle$$

$$- \sum_{i=1}^N y_i \alpha_i \left\langle \sum_{j=1}^N y_j \alpha_j x_j, x_i \right\rangle$$

$$+ C \sum_{i=1}^N \xi_i - b \sum_{i=1}^N y_i \alpha_i + \sum_{i=1}^N \alpha_i - \sum_{i=1}^N \alpha_i \xi_i - \sum_{i=1}^N \gamma_i \xi_i$$

$$(3) D(\xi, \alpha, \gamma) = \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N y_i y_j \alpha_i \alpha_j \langle x_i, x_j \rangle$$

$$+ \sum_{i=1}^N \alpha_i + \sum_{i=1}^N (C - \alpha_i - \gamma_i) \xi_i$$

dual SVM

$$\Rightarrow \left\{ \begin{array}{l} \min_{\alpha} \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N y_i y_j \alpha_i \alpha_j \langle x_i, x_j \rangle - \sum_{i=1}^N \alpha_i \\ \text{subject to} \quad \sum_{i=1}^N y_i \alpha_i = 0 \\ 0 \leq \alpha_i \leq C \text{ for all } i=1, \dots, N. \end{array} \right.$$

HW2. Q5: From the primal form SVM problem, derive the SVM dual problem. Additionally show that these optimization problems are equivalent to empirical risk minimization with the hinge loss plus ridge penalty.

$$\min_{w, b} \frac{1}{2} \|w\|^2, \quad y_n (\langle w, x_n \rangle + b) \geq 1, \text{ for all } n=1, \dots, N$$

hard

$$\min_{w, b, \xi} \underbrace{\frac{1}{2} \|w\|^2}_{\text{L2 norm}} + C \underbrace{\sum_{n=1}^N \xi_n}_{\text{regularization}}, \quad y_n (\langle w, x_n \rangle + b) \geq 1 - \xi_n$$

soft

$$\xi_n \geq 0$$

subject to

$$\forall i, \quad y_i (w^T x_i + b) \geq 1 - \xi_i$$

$$\forall i, \quad \xi_i \geq 0$$

$$\xi_i \geq 1 - y_i (w^T x_i + b)$$

find the weight vector w , and bias, b minimize

$$\frac{1}{2} \|w\|^2$$

and

$$C \sum_{i=1}^N \xi_i$$

L2 norm

regularization term

$$y_i (w^T x_i + b) \geq 1 - \xi_i, \quad \forall i, \quad \xi_i \geq 0$$

$$\xi_i \geq 1 - y_i (w^T x_i + b)$$

$$\xi_i \geq \max \{0, 1 - y_i (w^T x_i + b)\}$$

$$\Rightarrow \min_{w, b} \frac{1}{2} \|w\|^2 + C \sum_{n=1}^N \max \{0, 1 - y_i (\langle w, x_n \rangle + b)\}$$

error term

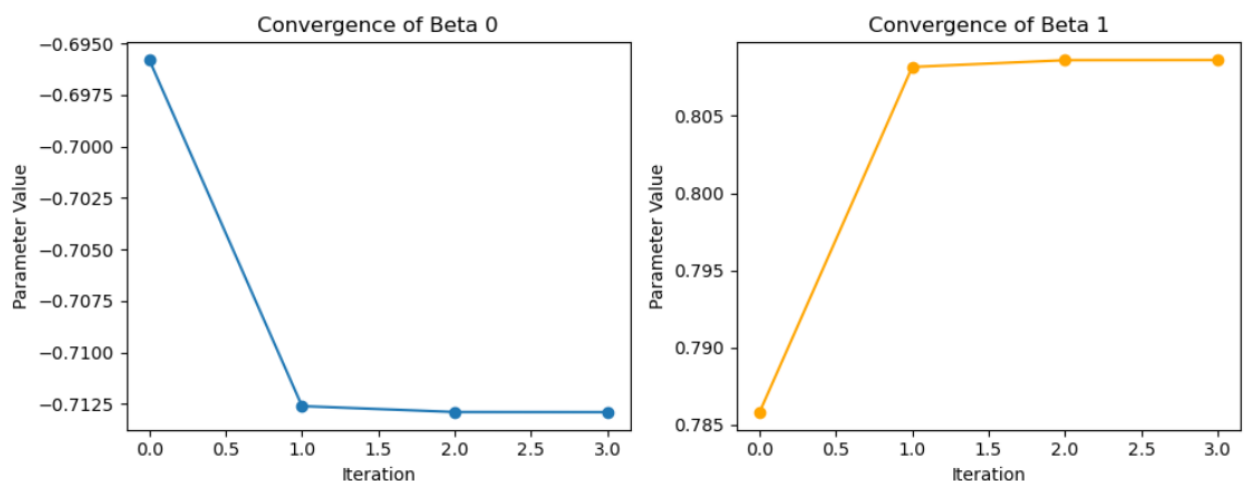
QUESTION 4

```
In [23]: import matplotlib.pyplot as plt
np.random.seed(0)
X = np.random.rand(100, 2)
y = np.random.randint(0, 2, 100)
# Initialize parameters
beta = np.zeros(2)
tolerance = 1e-6
max_iter = 100
iteration_values = []
parameter_values = []

# Newton's method
for i in range(max_iter):
    # predicted probabilities
    p = 1 / (1 + np.exp(-X.dot(beta)))
    z = X.dot(beta) + (y - p) / (p * (1 - p))
    # Calculate weighted design matrix and response
    X_w = X * (p * (1 - p))[:, np.newaxis]
    y_w = z * (p * (1 - p))
    # Update parameters
    beta_new = np.linalg.inv(X_w.T.dot(X_w)).dot(X_w.T).dot(y_w)
    # Check for convergence
    if np.all(np.abs(beta_new - beta) < tolerance):
        print("Converged after", i + 1, "iterations.")
        break
    beta = beta_new
    iteration_values.append(i)
    parameter_values.append(beta.copy())

plt.figure(figsize=(10, 4))
plt.subplot(1, 2, 1)
plt.plot(iteration_values, [params[0] for params in parameter_values], marker='o', label="Iteration")
plt.xlabel("Iteration")
plt.ylabel("Parameter Value")
plt.title("Convergence of Beta 0")
plt.subplot(1, 2, 2)
plt.plot(iteration_values, [params[1] for params in parameter_values], marker='o', label="Iteration")
plt.xlabel("Iteration")
plt.ylabel("Parameter Value")
plt.title("Convergence of Beta 1")
plt.tight_layout()
plt.show()
print("Converged parameters:", beta)
```

Converged after 5 iterations.



Converged parameters: [-0.71289898 0.80861772]

$$\frac{\partial^2 \ell(\beta)}{\partial \beta \partial \beta^T} = \sum_i \left[y_i (y_i p(x_i; \beta)) p(x_i; \beta) (1 - p(x_i; \beta)) (1 - 2p(x_i; \beta)) \right. \\ \left. - (1 - y_i) (1 / (1 - p(x_i; \beta))) p(x_i; \beta) (1 - p(x_i; \beta)) (1 - 2p(x_i; \beta)) \right]$$

$$\Rightarrow \left(\frac{\partial^2 \ell(\beta)}{\partial \beta \partial \beta^T} = - \sum (x_i x_i^T p(x_i; \beta) (1 - p(x_i; \beta))) \right)$$

Newton's

$$\beta^{(t+1)} = \beta^{(t)} - \left(\frac{\partial^2 \ell(\beta)}{\partial \beta \partial \beta^T} \right)^{-1} \frac{\partial \ell(\beta)}{\partial \beta}$$

Write the derived equations in matrix notation:

$$\begin{cases} Y: y_i, & X: [x_i]^{N \times (p+1)}, & p = p(x_i; \beta^t) \\ W: N \times N \text{ diagonal matrix} & \begin{bmatrix} p(x_i; \beta^t) (1 - p(x_i; \beta^t)) \\ \vdots \end{bmatrix} \end{cases}$$

$$\left(\frac{\partial \ell(\beta)}{\partial \beta} = X^T (y - p) \right)$$

$$\left(\frac{\partial^2 \ell(\beta)}{\partial \beta \partial \beta^T} = -X^T W X \right)$$

$$\beta^{t+1} = \beta^t + (X^T W X)^{-1} X^T (y - p)$$

$$= (X^T W X)^{-1} X^T W (X \beta^t + W^{-1} (y - p))$$

$$\beta^{t+1} = (X^T W X)^{-1} X^T W y$$

//

$$\frac{d\ell}{d\beta} = \sum_{i=1}^N \left[y_i (1 - p(x_i; \beta)) x_i - (1 - y_i) p(x_i; \beta) x_i \right]$$

$$\frac{\partial \ell}{\partial \beta} = \sum_{i=1}^N \left[y_i x_i - \cancel{y_i p(x_i; \beta) x_i} - p(x_i; \beta) x_i + \cancel{y_i p(x_i; \beta) x_i} \right]$$

$$(*) \quad \frac{\partial \ell}{\partial \beta} = \sum_{i=1}^N x_i (y_i - p(x_i; \beta))$$

Step 2: $\frac{\partial^2 \ell(\beta)}{\partial \beta \partial \beta^T}$

first $\frac{\partial^2}{\partial \beta \partial \beta^T} (y_i \log(p(x_i; \beta))) = -y_i p(x_i; \beta) (1 - p(x_i; \beta)) (x_i x_i^T)$

second $\frac{\partial^2}{\partial \beta \partial \beta^T} ((1 - y_i) \log(1 - p(x_i; \beta))) = -(1 - y_i) (1 - p(x_i; \beta)) p(x_i; \beta) (x_i x_i^T)$

$$\sigma'(t) = \frac{1}{1 + e^{-t}}$$

$$\sigma' = \sigma(t) (1 - \sigma(t))$$

$$\sigma'' = \sigma'(t) (1 - \sigma(t)) - \sigma(t) \sigma'(t)$$

$$= \sigma'(t) (1 - \sigma(t)) (1 - 2\sigma(t))$$

$$t \rightarrow \beta^T x_i$$

$$\frac{\partial^2 \ell(x_i; \beta)}{\partial \beta \partial \beta^T} = p(x_i; \beta) (1 - p(x_i; \beta)) (1 - 2p(x_i; \beta)) x_i x_i^T$$

HW 2. Q4: Derive an algorithm to fit logistic regression using Newton's method. Show that this approach can be written as IRWLS.

Logistic Regression:

$$P_r(G=k|X=x) = \frac{1}{1 + \sum_{l=1}^{K-1} e^{(\beta_{0l} + \beta_l^T x)}}$$

$$P_r(G=k|X=x) = p_k(X; \theta).$$

Log likelihood for N observations:

$$l(\theta) = \sum_{i=1}^N \log p_{g_i}(x_i; \theta)$$

Two class:

$$(A) \quad l(\beta) = \sum_{i=1}^N \left[y_i \log p(x_i; \beta) + (1-y_i) \log (1-p(x_i; \beta)) \right]$$

(B) Newton-Rapson algorithm

$$\beta^{(t+1)} = \beta^{(t)} - \left(\overset{\text{step 2}}{\frac{\partial^2 l(\beta)}{\partial \beta \partial \beta^T}} \right)^{-1} \overset{\text{step 1}}{\frac{\partial l(\beta)}{\partial \beta}}$$

Step 1: $\frac{\partial}{\partial \beta} l(\beta)$

first term

$$\frac{\partial}{\partial \beta} \left(y_i \log(p(x_i; \beta)) \right) = y_i \frac{1}{p(x_i; \beta)} \cdot \frac{d}{d\beta} p(x_i; \beta) \quad (1)$$

second term

$$\frac{\partial}{\partial \beta} \left((1-y_i) \log(1-p(x_i; \beta)) \right) = (1-y_i) \frac{1}{1-p(x_i; \beta)} \cdot \frac{d}{d\beta} (1-p(x_i; \beta)) \quad (2)$$

$$(1) \quad \frac{d}{d\beta} p(x_i; \beta) = p(x_i; \beta)(1-p(x_i; \beta)) \frac{d}{d\beta} (\beta^T x_i)$$

$$(2) \quad \frac{d}{d\beta} (1-p(x_i; \beta)) = -p(x_i; \beta)(1-p(x_i; \beta)) \frac{d}{d\beta} (\beta^T x_i)$$

Sd:

The Variance :

$$\frac{1}{N} \sum_{i=1}^N z_{i1}^2 = \frac{1}{N} \sum_{i=1}^N w_1^T x_i x_i^T w_1 = w_1^T \hat{\Sigma} w_1$$

The mean :

$$\hat{\mu}_k = \frac{1}{N} \sum_{i \in \omega_k} \tilde{x}_i = w^T \hat{\mu}_k$$

LDA

$$\hat{\Sigma}_w^{-1} \hat{\Sigma}_b w = \lambda w, \quad \hat{\Sigma}_b = (\hat{\mu}_1 - \hat{\mu}_2)(\hat{\mu}_1 - \hat{\mu}_2)^T$$

$$\Rightarrow w = \lambda^{-1} \hat{\Sigma}_w^{-1} \hat{\Sigma}_b w = \hat{\Sigma}_w^{-1} (\hat{\mu}_1 - \hat{\mu}_2) (\hat{\mu}_1 - \hat{\mu}_2)^T w$$

$$w = C_1 \hat{\Sigma}_w^{-1} (\hat{\mu}_1 - \hat{\mu}_2)$$

(1) So, Eigenvector w and LDA decision boundaries are parallel

FDA :

$$-\frac{1}{2} \tilde{\mu}_1^T \tilde{\Sigma}_w^{-1} \tilde{\mu}_1 + \frac{1}{2} \tilde{\mu}_2^T \tilde{\Sigma}_w^{-1} \tilde{\mu}_2 + \tilde{x}^T \tilde{\Sigma}_w^{-1} (\tilde{\mu}_1 - \tilde{\mu}_2) = 0$$

$$\tilde{x}^T \tilde{\Sigma}_w^{-1} (\tilde{\mu}_1 - \tilde{\mu}_2) = \tilde{x}^T w w^T \hat{\Sigma}_w^{-1} w w^T (\hat{\mu}_1 - \hat{\mu}_2)$$

$$w = C_2 \tilde{x}^T w$$

(2) FDA decision boundary is also parallel to w,

* So, LDA and FDA decision boundaries are parallel

$$\sum_{i=1}^k \hat{\Sigma}_i^{-1} \hat{\Sigma}_i = V \Lambda V^T$$

top eigenvector

if $k=2$;

$$p(y=1 | x, \theta) = \frac{e^{\beta_1^T x + \theta_1}}{e^{\beta_1^T x + \theta_1} + e^{\beta_0^T x + \theta_0}} = \frac{1}{1 + e^{(\beta_0 - \beta_1)^T x + (\theta_0 - \theta_1)}}$$

$$= \text{sigm}((\beta_1 - \beta_0)^T x + (\theta_1 - \theta_0))$$

$$\gamma_1 - \gamma_0 = -\frac{1}{2} \mu_1^T \Sigma^{-1} \mu_1 + \frac{1}{2} \mu_0^T \Sigma^{-1} \mu_0 \log(\mu_1 / \mu_0)$$

LOA: $-\frac{1}{2} \hat{\mu}_1^T \hat{\Sigma}_1^{-1} \hat{\mu}_1 + \frac{1}{2} \mu_2^T \hat{\Sigma}_2^{-1} \hat{\mu}_2 + \underbrace{x^T \hat{\Sigma}_w^{-1} (\hat{\mu}_1 - \hat{\mu}_2)}_{=0} = 0$

Projected data $w_i \in \mathbb{R}^r, \tilde{z}_i \in \mathbb{R}^N$

$$J(w_1, z_1) = \frac{1}{N} \sum_{i=1}^N \|x_i - z_i w_1\|^2 = \frac{1}{N} \sum_{i=1}^N (x_i - z_i w_1)^T (x_i - z_i w_1)$$

$$= \frac{1}{N} \sum_{i=1}^N [x_i^T x_i - 2 z_i w_1^T x_i + z_i^T w_1]$$

$$= \frac{1}{N} \sum_{i=1}^N [x_i^T x_i - 2 z_i w_1^T x_i + z_i^2] \quad \left\{ w_1^T w_1 = 1 \right\}$$

$$\frac{\partial}{\partial z_{i1}} J(w_1, z_1) = \frac{1}{N} [-2 w_1^T x_i + 2 z_{i1}] = 0 \Rightarrow z_{i1} = w_1^T x_i$$

$$J(w_1) = \frac{1}{N} \sum_{i=1}^N [x_i^T x_i - z_{i1}^2] = \text{const} - \frac{1}{N} \sum_{i=1}^N z_{i1}^2$$

$$\text{var}[z_i] = \mathbb{E}[z_i^2] - (\mathbb{E}[z_i])^2 = \frac{1}{N} \sum_{i=1}^N z_{i1}^2 - 0$$

$$\mathbb{E}[z_{i1}] = \mathbb{E}[x_i^T w_1] = \mathbb{E}[x_i]^T w_1 = 0$$

$$\arg \min_{w_1} J(w_1) = \arg \max_{w_1} \text{var}[z_i]$$

HW.2. Q2: Prove that under certain assumptions, FDA is equivalent to LDA:

Generative Classifier, $p(x|y=c)$; class conditional ²³ density

$$p(y=c|x,\theta) = \frac{p(y=c|\theta)p(x|y=c,\theta)}{\sum_c p(y=c|\theta)p(x|y=c,\theta)}$$

Quadratic discriminant analysis:

$$p(y=c|x,\theta) = \frac{\pi_c |2\pi \Sigma_c|^{-1/2} e^{-\frac{1}{2}(x-\mu_c)^T \Sigma_c^{-1}(x-\mu_c)}}{\sum_c \pi_c |2\pi \Sigma_c|^{-1/2} e^{-\frac{1}{2}(x-\mu_c)^T \Sigma_c^{-1}(x-\mu_c)}}$$

LDA:

$$\Sigma_c = \Sigma,$$

$$\begin{aligned} p(y=c|x,\theta) &\propto \pi_c e^{-\left[\mu_c^T \Sigma^{-1}x - \frac{1}{2}x^T \Sigma^{-1}x - \frac{1}{2}\mu_c^T \Sigma^{-1}\mu_c\right]} \\ &= e^{-\left[\mu_c^T \Sigma^{-1}x - \frac{1}{2}\mu_c^T \Sigma^{-1}\mu_c + \log \pi_c\right]} e^{-\frac{1}{2}x^T \Sigma^{-1}x} \end{aligned}$$

(i) $x^T \Sigma^{-1}x$ is independent of c ;

$$(ii) \gamma_c = -\frac{1}{2}\mu_c^T \Sigma^{-1}\mu_c + \log \pi_c$$

$$(iii) \beta_c = \Sigma^{-1}\mu_c$$

$$\Rightarrow p(y=c|x,\theta) = \frac{e^{\beta_c^T x + \gamma_c}}{\sum_c e^{\beta_c^T x + \gamma_c}} = S(\eta)_c$$

$$\text{where } \eta = [\beta_1^T x + \gamma_1, \dots, \beta_C^T x + \gamma_C]$$

$$S(\eta)_c = \frac{e^{\eta_c}}{\sum_{c=1}^C e^{\eta_c}}, \text{ softmax function}$$

$$\log p(y=c|x,\theta) \Rightarrow$$

$$\left\{ \begin{aligned} p(y=c|x,\theta) &= p(y=c|x,\theta) \\ \beta_c^T x + \gamma_c &= \beta_{c'}^T x + \gamma_{c'} \\ x^T(\beta_{c'} - \beta) &= \gamma_{c'} - \gamma_c \end{aligned} \right\}$$

Gaussian Process Regression

Noisy training data

$$y_i = f(x_i) + \epsilon, \text{ where } \epsilon \sim N(0, \sigma^2)$$

(i) $f(x) \sim N(m(x), k(x, x'))$; Gaussian Process prior

(ii) $y|x, f(x) \sim N(f(x), \sigma^2)$; likelihood

$$\begin{bmatrix} f(x) \\ y \end{bmatrix} \sim N \left(\begin{pmatrix} m(x) \\ m(x) \end{pmatrix}, \begin{pmatrix} k(x, x') & k(x, x') \\ k(x', x) & k(x', x) + \sigma^2 \end{pmatrix} \right)$$

Conditional distribution of $f(x)$ given y

$$\mathbb{E}[y'|y] = \mu$$

$$\mu(x) = m(x) + \frac{k(x, x)}{k(x, x) + \sigma^2 \mathbf{I}} (y - m(x))$$

$$\Sigma(x) = k(x, x) - \frac{k(x, x) k(x, x)}{k(x, x) + \sigma^2 \mathbf{I}}$$

$$y'|y \sim N \left(m(x) + k(x, x)^T [k(x, x) + \sigma^2 \mathbf{I}]^{-1} (y - m(x)), \right. \\ \left. k(x, x) - k(x, x)^T [k(x, x) + \sigma^2 \mathbf{I}]^{-1} k(x, x) \right)$$

MAP estimate is the mean of Posterior Gaussian.

$$\mathbb{E}[y'|y] = \mu = \underbrace{m(x)}_{\text{mean of testing}} + \underbrace{k(x, x) [k(x, x) + \sigma^2 \mathbf{I}]^{-1} (y - m(x))}_{\text{mean of training}}$$

when centered

$$*** \mu = k(x, x)^T (k(x, x) + \sigma^2 \mathbf{I})^{-1} = \hat{f}(x) ***$$

Kernel Ridge estimate

HW2-Q1: Gaussian Process Regression. Show that the posterior mean and mode of Gaussian Process regression (MAP estimator) is the same as kernel ridge estimator.

Kernel Ridge Regression

Given a training data set $\{(x_i, y_i)\}_{i=1}^N$, $y_i \rightarrow$ target
 $x_i \rightarrow$ input features.

$$f(x) = \sum_{i=1}^N \beta_i K(x, x_i) \quad ; \text{ Kernel Ridge Regression}$$

K_i : Kernel β_i : coefficient

$$\mathcal{L} = \|y - K(x, x_i) \beta\|_2^2 + \lambda \beta^T K(x, x_i) \beta$$

$$\nabla_{\beta} \mathcal{L} = 0, \text{ gives } \hat{\beta}$$

$$\nabla \left(\|y - K(x, x_i) \beta\|_2^2 + \lambda \beta^T K(x, x_i) \beta \right) = 0$$

$$-2K^T(y - K(x, x_i)\beta) + 2\lambda I \beta = 0$$

$$-2K^T(y - K\beta) + 2\lambda I \beta = 0$$

$$K^T K \beta - K^T y + \lambda I \beta = 0$$

$$K^T K \beta + \lambda I \beta = K^T y$$

$$(K^T K + \lambda I) \beta = K^T y$$

$$\beta = (K^T K + \lambda I)^{-1} K^T y$$

also

$$\beta^* = (K + \lambda I)^{-1} y$$

$$f(x) = \sum_{i=1}^N \beta_i K(x, x_i)$$

$$\hat{f}(x) = K(x)^T (K + \lambda I)^{-1} y$$